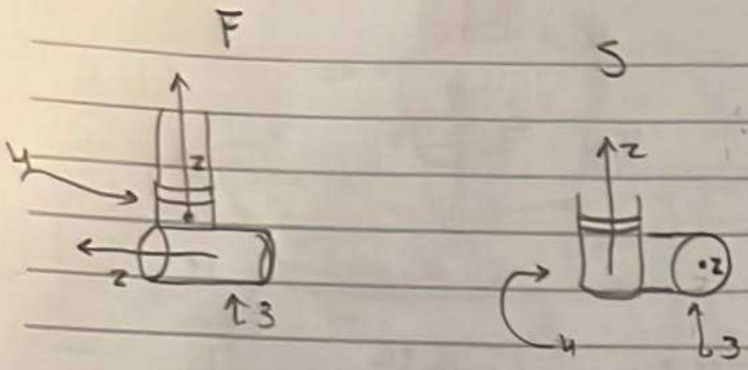


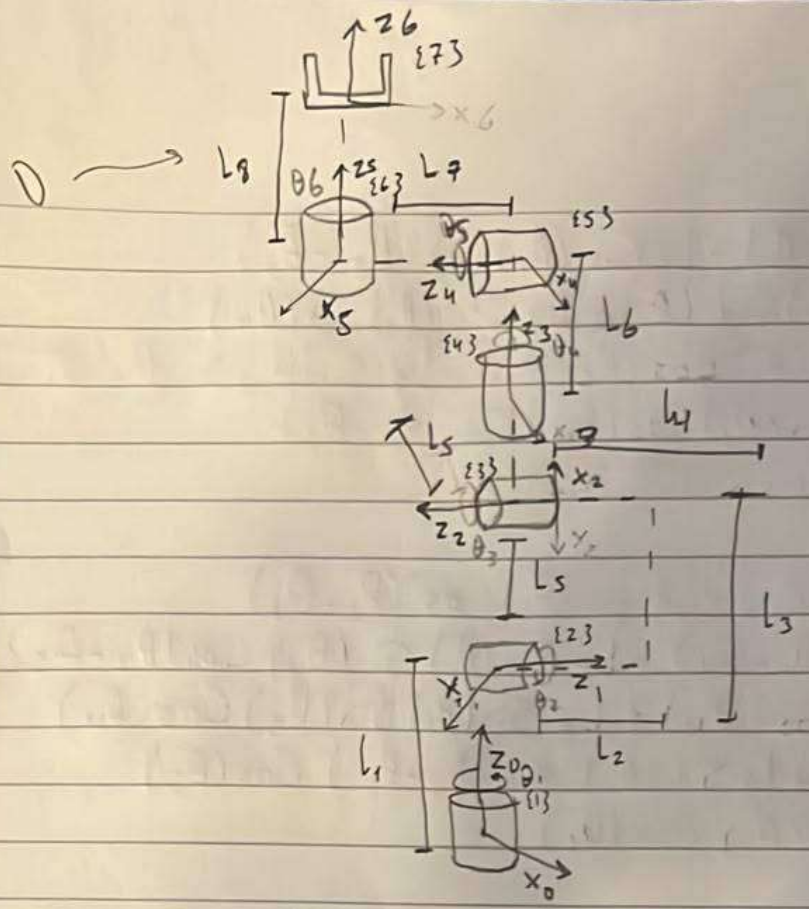
$$
{}^{0}T_{1} = \begin{bmatrix} C_{\theta_1} & 0 & -S_{\theta_1} & 0 \\ S_{\theta_1} & C_{\theta_1} & +C_{\theta_1} & 0 \\ 0 & -1 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{1}T_{2} = \begin{bmatrix} C_{\theta_2} & +S_{\theta_2} & 0 & L_3 C_{\theta_2} \\ S_{\theta_2} & -C_{\theta_2} & 0 & L_3 S_{\theta_2} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{2}T_{3} = \begin{bmatrix} C_{\theta_3} & 0 & S_{\theta_3} & L5\, C_{\theta_3} \\ S_{\theta_3} & 0 & -C_{\theta_3} & L5\, S\theta 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{3}T_{4} = \begin{bmatrix} C_{\theta 4} & 0 & S_{\theta 4} & 0 \\ S_{\theta 4} & 0 & -C_{\theta 4} & 0 \\ 0 & 1 & 0 & L6 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{4}T_{5} = \begin{bmatrix} C_{\theta 5} & 0 & -S_{\theta 5} & 0 \\ S_{\theta 5} & 0 & +C_{\theta 5} & 0 \\ 0 & -1 & 0 & L7 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$



i	θ_i	d_i	a_i	α_i
1	θ_1	L_1	0	$-\frac{\pi}{2}$
2	$\theta_2(-\pi+\theta_2)$	0	L_3	π
3	θ_3	0	L_5	$+\frac{\pi}{2}$
4	θ_4	L_6	0	$+\pi/2$
5	θ_5	L_7	0	$-\frac{\pi}{2}$
6	θ_6	L_8	0	0

$$
{}^{5}T_6 = \begin{bmatrix} C_{\theta_6} & -S_{\theta_6}C_0 & S_{\theta_6}S_0 & 0 \\ S_{\theta_6} & +S_{\theta_6}C_0 & -C_{\theta_6}S_0 & 0 \\ 0 & S_0 & C_0 & L_8 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{0}T_6 = \begin{bmatrix} a_{11} & a_{12} & a_{13} & X \\ a_{21} & a_{22} & a_{23} & Y \\ a_{31} & a_{32} & a_{33} & Z \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$X = -L_3 \cos(\theta_1)\cos(\theta_2) - L_5 \cos(\theta_1)\cos(\theta_2 - \theta_3)$$
$$+ L_6 \sin(\theta_2 - \theta_3)\cos(\theta_1) - L_7 \sin(\theta_1)\cos(\theta_4)$$
$$- L_7 \sin(\theta_4)\cos(\theta_1)\cos(\theta_2 - \theta_3) - L_8 \sin(\theta_1)\sin(\theta_4)\sin(\theta_5)$$
$$+ L_8 \sin(\theta_5)\cos(\theta_1)\cos(\theta_4)\cos(\theta_2 - \theta_3)$$
$$+ L_8 \sin(\theta_2 - \theta_3)\cos(\theta_1)\cos(\theta_5)$$

$$Y = -L_3 \sin(\theta_1)\cos(\theta_2) - L_5 \sin(\theta_1)\cos(\theta_2 - \theta_3)$$
$$+ L_6 \sin(\theta_1)\sin(\theta_2 - \theta_3) - L_7 \sin(\theta_1)\sin(\theta_4)\cos(\theta_2 - \theta_3)$$
$$+ L_7 \cos(\theta_1)\cos(\theta_4) + L_8 \sin(\theta_1)\sin(\theta_5)\cos(\theta_4)$$
$$\cos(\theta_2 - \theta_3) + L_8 \sin(\theta_1)\sin(\theta_2 - \theta_3)\cos(\theta_5)$$
$$+ L_8 \sin(\theta_4)\sin(\theta_5)\cos(\theta_1)$$

$$Z = L_1 + L_3 \sin(\theta_2) + L_5 \sin(\theta_2 - \theta_3) + L_6 \cos(\theta_2 - \theta_3)$$
$$+ L_7 \sin(\theta_4)\sin(\theta_2 - \theta_3) - L_8 \sin(\theta_5)\sin(\theta_2 - \theta_3)\cos(\theta_4)$$
$$+ L_8 \cos(\theta_5)\cos(\theta_2 - \theta_3)$$